



When Space Meets Time

Statistical AI for Public Health

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Outline

- My Journey
- Introduction
- Spatiotemporal AQI data: EVT meets GNNs
- Spatiotemporal epidemic data: Deep Generative Modeling
- Future direction

My Journey so far



MS from Indian Statistical Institute (2014-16)
PhD from ISI Kolkata (2016-2020)
B.G. Raghavendra Memorial Award

...

2016

...

2021

...

2026

...

- ✓ ACM IKDD Best Paper Award
- ✓ Statistical Machine Learning
- ✓ Direct Industrial Applications

- ✓ 200+ hours/yr of classroom teaching
- ✓ Grants from SAFIR, SCAI, Mphasis
- ✓ Thesis Supervision: PhD (4) MS (4)



Joined A/Prof. at SUAD in 2021
Promoted to Associate Prof. in 2024
Developed BS Math courses at SUAD



HDR in Math at SU Paris (2026)
Adjunct A/Prof at NUS Singapore
Investigator at CIDSAI, NYUAD



Funders:



Introduction

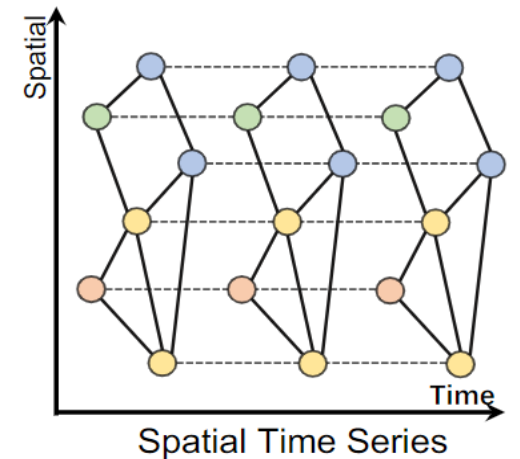
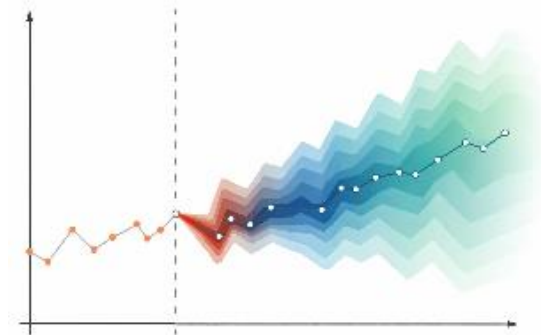
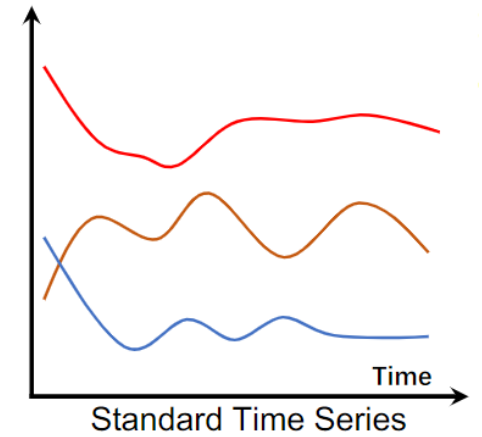
Time and Space

Time series is a set of observations - each one being recorded at a specific time.

Forecasting is estimating how the sequence of observations will continue into the future.

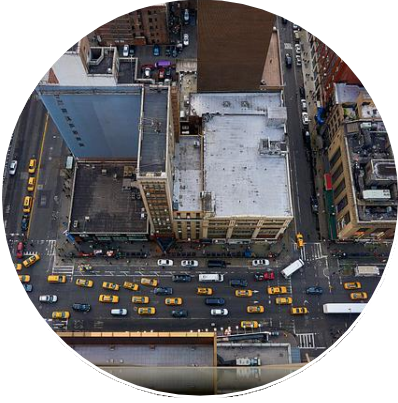
Spatio-temporal data is a type of data that combines spatial and temporal dimensions, capturing how measurements or observations vary across different locations over time.

Spatiotemporal data can be modeled as **graph-structured data**, particularly when the data involves complex relationships, such as sensor networks, traffic flow, or social interactions.



Examples

- A lot of real-world data does not “live” on grids



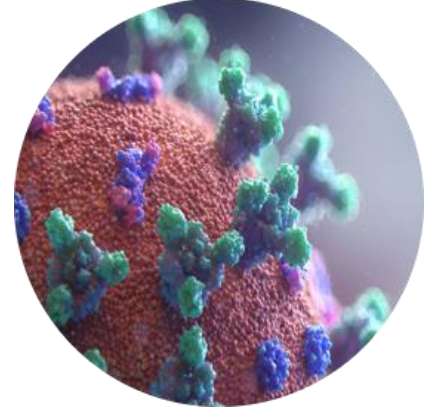
Traffic monitoring



Weather monitoring



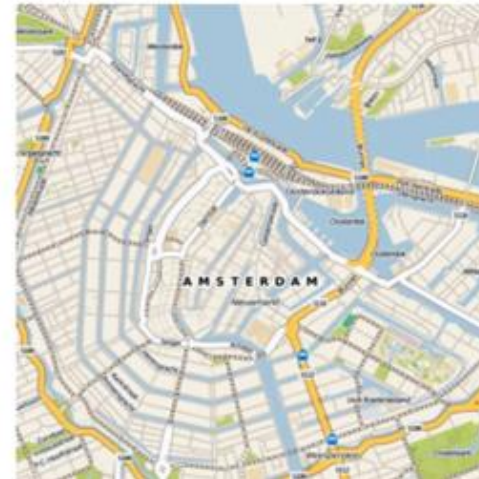
Social Interactions



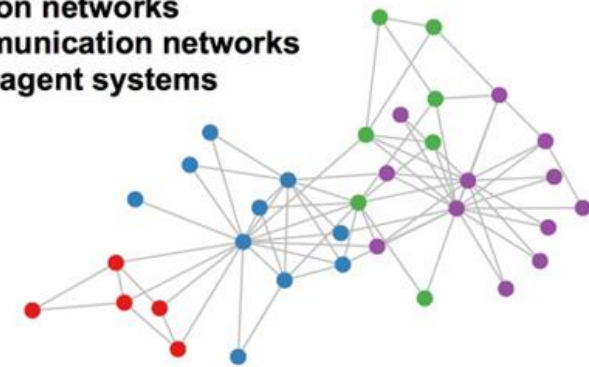
Epidemics

- Graphs are the **main** modality of data we receive from **nature**

Standard **CNN** and **RNN** architectures don't work on this data



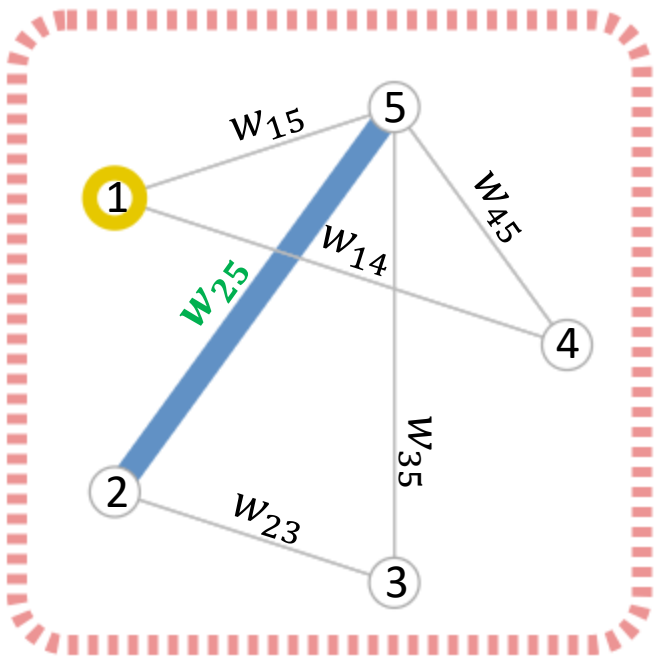
Social networks
Citation networks
Communication networks
Multi-agent systems



Road maps

What is a Graph?

- **Graph** represents the relations (edges) between a collection of entities (nodes) quantified using weights.
- **Graph $G(V, E, W)$** : V = nodes, E = edges (link node pairs), W = weights associated with edges.
- **Adjacency matrix (A)** of a graph is a sparse matrix with non-zero elements denoting graph weights.
- The elements of the matrix indicate whether pairs of nodes are **adjacent** or not in the graph.



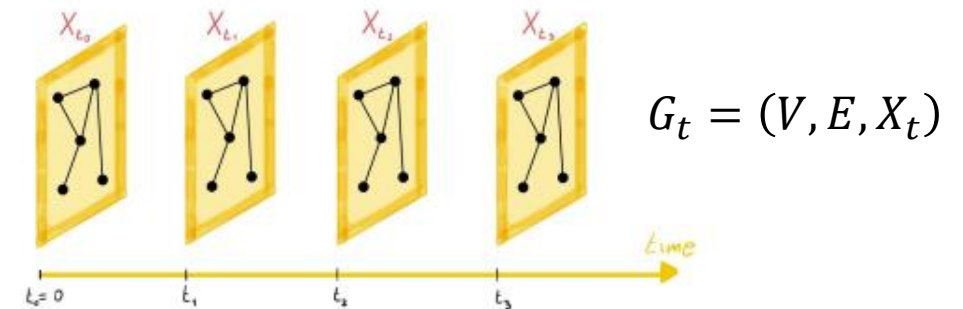
Vertex (or node)

Edges (or link)

Weights

**Global context
(or master node)**

Fig: Attributes of undirected weighted graph

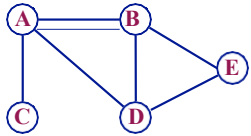


Spatio-Temporal Graph

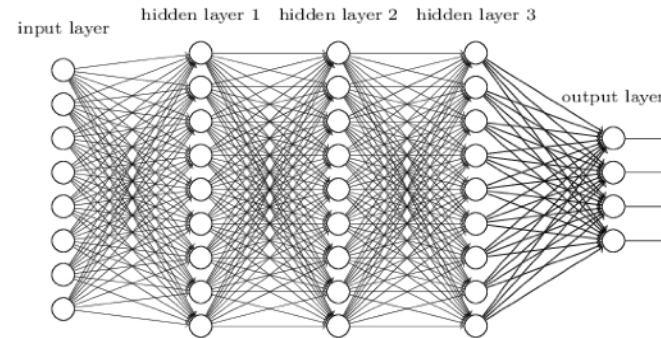
- Topology is fixed, but features of nodes (X_t) change over time
- (Usually) observed at regular intervals
- Examples: Air pollution level

A naïve approach

- Take adjacency matrix: $\mathbf{A}$ and feature matrix: $\mathbf{X}$
- Concatenate them: $\mathbf{X}_{\text{in}} = [\mathbf{A}, \mathbf{X}]$ **Graph:** $G = (\mathcal{V}, \mathcal{E})$
- Feed them into deep (fully connected) neural net



	A	B	C	D	E	Feat	
A	0	1	1	1	0	1	0
B	1	0	0	1	1	0	0
C	1	0	0	1	0	0	1
D	1	1	1	0	1	1	1
E	0	1	0	1	0	1	0



? We need weight sharing!

Possible Solution:

Deep Learning on graphs or
Graph Neural (Convolutional) Networks

Problems:

- Huge number of parameters
- Re-train if graph changes
- Can't exploit neighborhood info

Graph Neural Networks (GNNs)

Notation: $\mathcal{G} = (\mathbf{A}, \mathbf{X})$

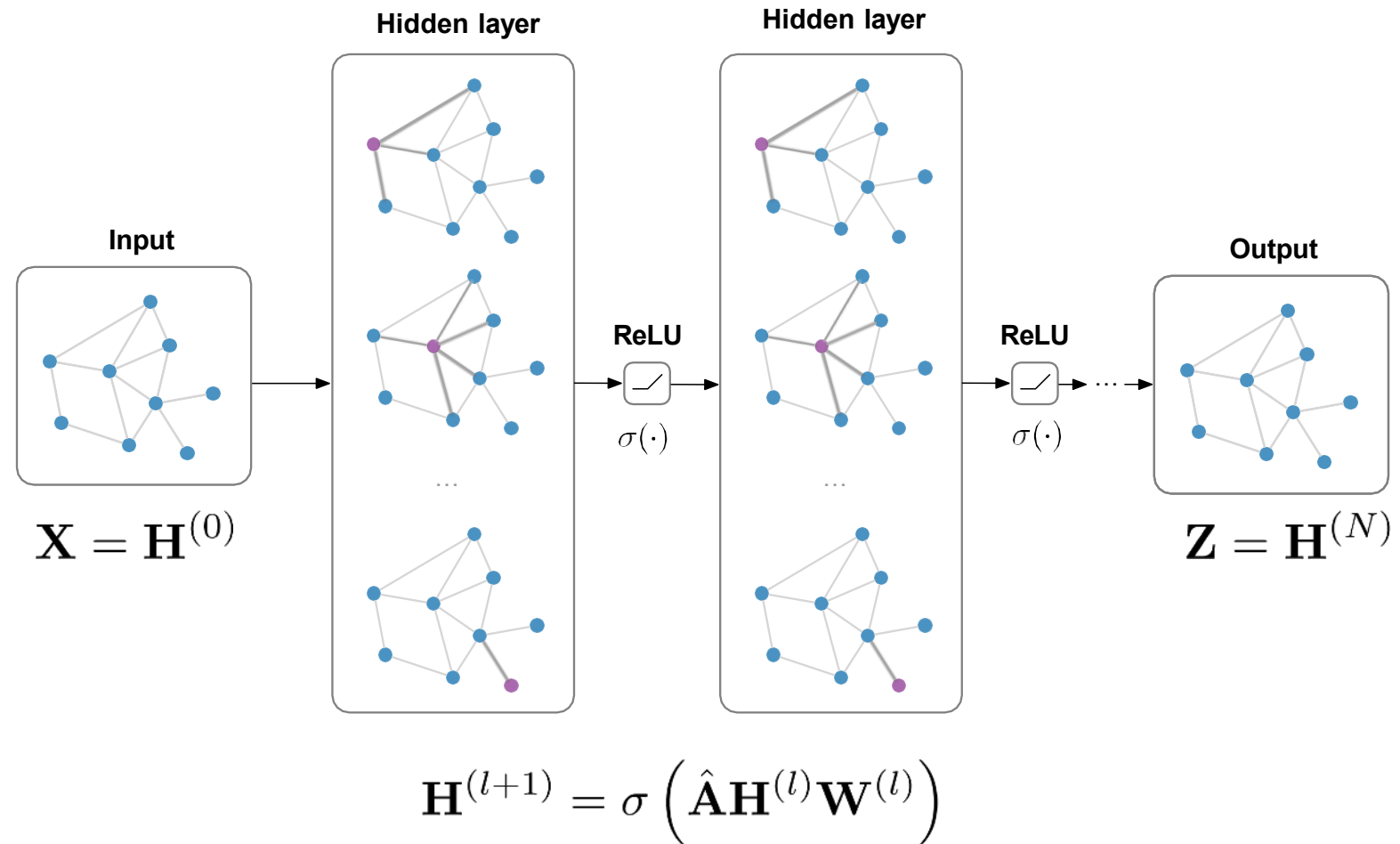
- Adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$
- Feature matrix $\mathbf{X} \in \mathbb{R}^{N \times E}$

N = number of **nodes** in the graph

E = number of **features per node**.

Main Idea: Pass messages between pairs of nodes and agglomerate

Alternative Interpretation: Pass messages between nodes to refine node (and possibly edge) representations



Neural message passing in GNNs

- GNNs utilize **neural message passing network** to exchange information amongst neighboring nodes and influence each other's updated embeddings as follows:
 - For each node in the graph, all the neighboring node embeddings (**messages**) are gathered.
 - All messages are aggregated via an **aggregate** function (like sum).
 - All pooled messages are passed through an **update function**, usually a learned neural network.

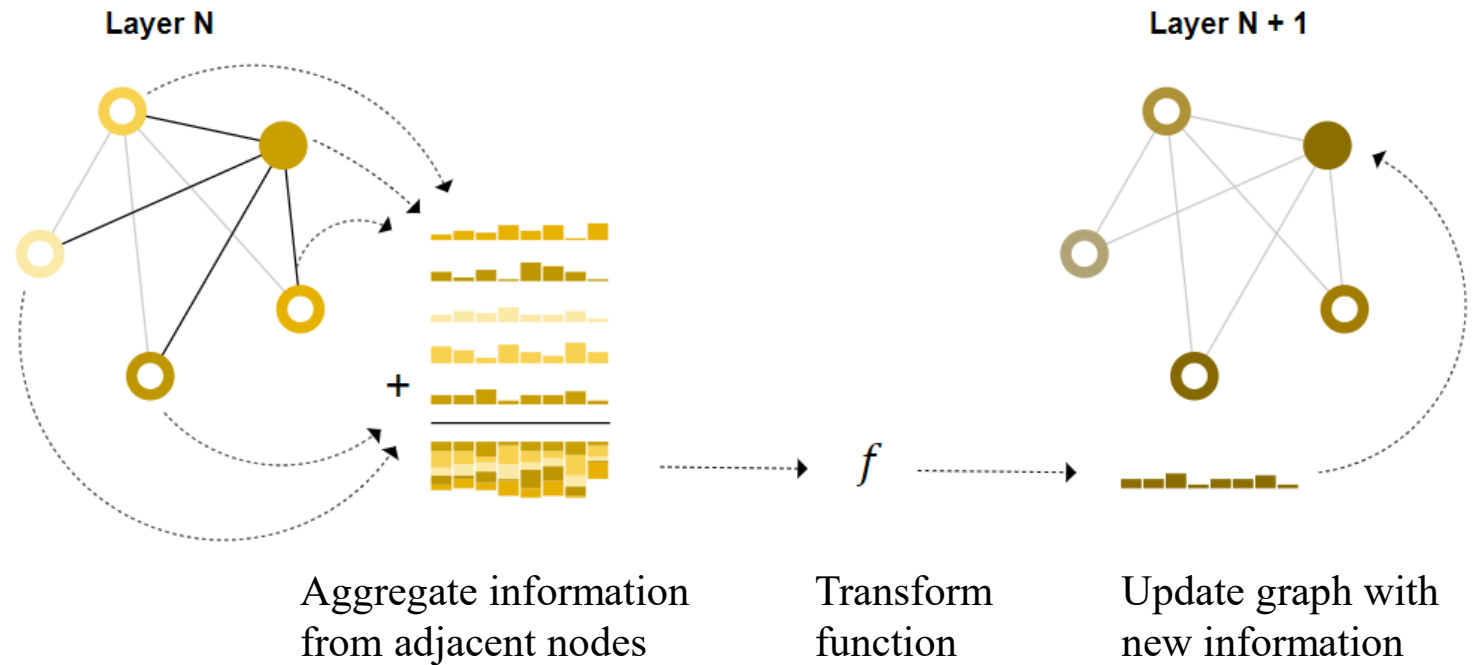


Fig: Updating the highlighted node embedding using adjacent nodes through neural message passing network.

Prediction task using GNNs

- The transformed graph embedded contains node information through GNN.
- For the prediction task we apply a [regressor or classifier](#) on each of the learned node representations.

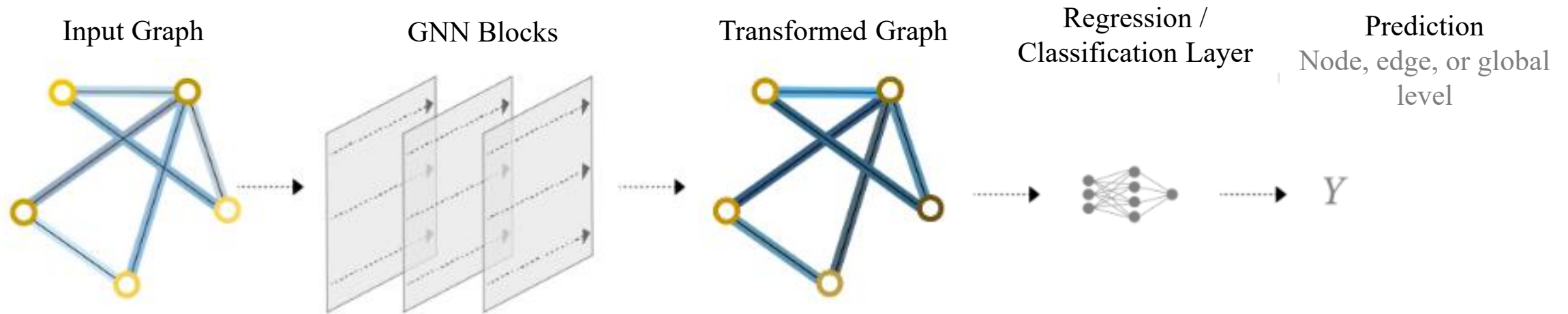


Fig: An end-to-end prediction-task using GNN.

- A GNN is the general family of neural networks for graph data, while a [Graph Convolutional Networks](#) (GCNs) is a specific GNN that aggregates neighboring nodes using normalized graph convolution.

Spatiotemporal AQI data: EVT meets GNN

*Journal of the Royal Statistical Society Series A:
Statistics in Society*, 2026, **00**, 1–51
<https://doi.org/10.1093/jrsssa/qnag010>



Original Article

E-STGCN: extreme spatio-temporal graph convolutional networks for air quality forecasting

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Air Quality: A Global Concern

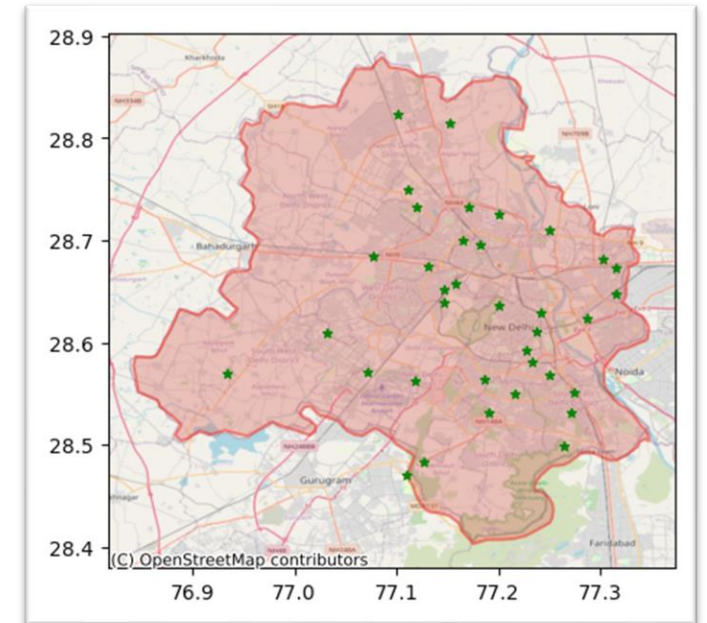


- Emission of particulate matter and harmful gases in the air from industrialization degrades air quality levels leading to harmful health hazards.
- WHO has set National Ambient Air Quality Standards (NAAQS) for major air pollutants to protect the public health and environment.
- Despite several efforts, Delhi (Capital of India) remains the most polluted capital city in the world as of early 2024.

Air Quality Data of Delhi

- In efforts to monitor and reduce the air pollution levels, the Central Pollution Control Board (CPCB) in Delhi has installed a network of forty ground-based sensors at different strategic locations.
- The sensors collect near real-time (hourly) data on the concentration levels of major air pollutants, e.g., PM_{10} , $\text{PM}^*_{2.5}$, NO_2 .
- This timely information empowers the pollution control management system, facilitating informed decision-making to reduce pollution levels.

* particles that are 2.5 microns or less in diameter



Delhi Air Quality Data Characteristics

Daily pollutant concentration levels of $\text{PM}_{2.5}$, PM_{10} , NO_2 were obtained from 40 ground-based sensors of Delhi's CPCB during 1st January 2019 till 31st December 2023.

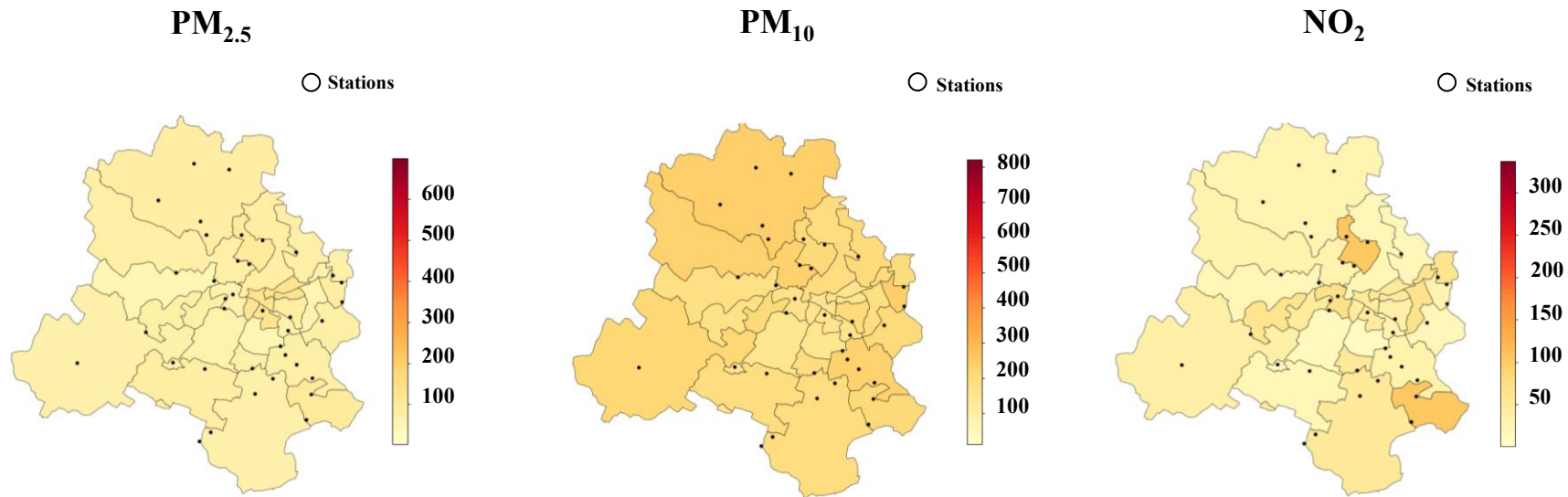


Fig: Evolution of Pollution Concentration recorded in Delhi during 2019 – 2022*.

- ✓ Nonlinear
- ✓ Non-stationary
- ✓ Long-term dependency
- ✓ Seasonality
- ✓ Extreme events

*NAAQS Thresholds for $\text{PM}_{2.5}$ is $60 \mu\text{g}/\text{m}^3$, PM_{10} is $100 \mu\text{g}/\text{m}^3$, and NO_2 is $80 \mu\text{g}/\text{m}^3$ ($\mu\text{g}/\text{m}^3$ - micrograms per cubic meter air)

Extreme Value Theory

- In classical statistics: focus on average behavior of stochastic process (Central Limit Theorem)
- In extreme value theory: focus on extreme and rare events (Fisher-Tippett Theorem)
- Block Maxima Method

$$M_n = \max\{X_1, \dots, X_n\}$$

M_n follows a Generalized Extreme Value distribution

- Peak over Threshold (PoT)

$$\{X_i - \tau \mid X_i > \tau\}$$

very large threshold τ follow a Generalized Pareto distribution (GPD)

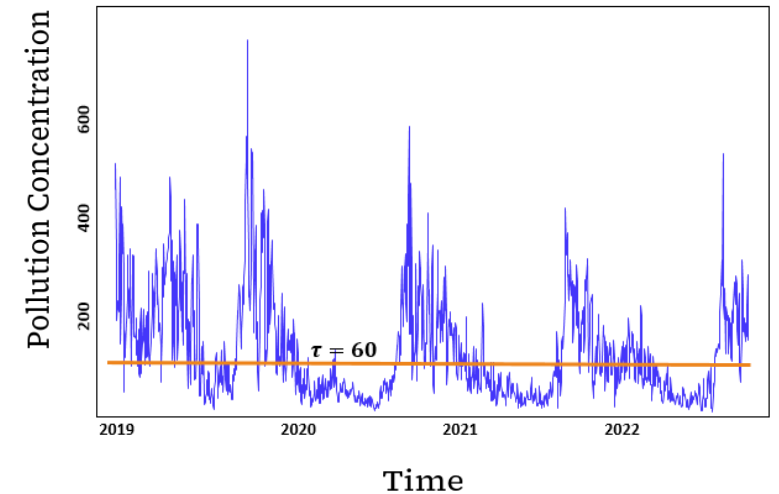


Fig: PoT method on PM_{2.5} level in Alipur, Delhi

Extreme Values in AQI Data

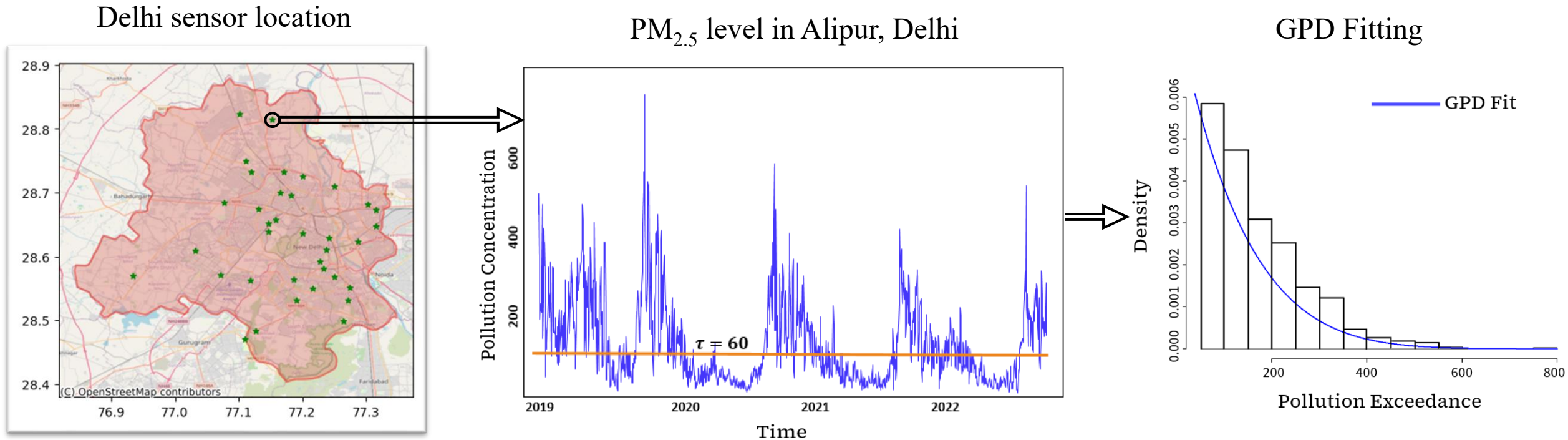


Fig: PoT analysis of PM_{2.5} concentration level recorded daily during 2019 – 2022 at Alipur, Delhi

- In Delhi, the air pollution level increases with the onset of winter due to the land-locked geographical location and low temperature restricting the dispersion of the pollutants (Extreme measurements arise).
- How can we accurately forecast AQIs for Delhi in the presence of extremes?

E-STGCN Framework

- **Input:** Pollutant concentrations across N sensors at timestamp t , $\mathbf{X}_t = \{X_t^1, \dots, X_t^N\} \in \mathbb{R}^N$; $t = 1, \dots, T$.
- **Output:** q (≥ 1) steps ahead forecast $\hat{\mathbf{X}}_{T+1}, \dots, \hat{\mathbf{X}}_{T+q}$ based on T historical observations.
- **Procedure:**
 - Design a spatiotemporal graph with nodes (sensor locations) and edges (interconnections).
 - The input can be represented as $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_T\}$ with $\mathcal{G}_t = \{\mathbf{X}_t, A\}$ being the graph at time t and $A \in \mathbb{R}^{N \times N}$ being the static adjacency matrix.
 - Compute $A = [a_{ij}]_{N \times N}$ using the **weighted Haversine distance** as

$$d_{ij} = 2R \sin^{-1} \left[\sqrt{\sin^2 \left(\frac{\phi^i - \phi^j}{2} \right) + \cos(\phi^i) \cos(\phi^j) \sin^2 \left(\frac{\lambda^i - \lambda^j}{2} \right)} \right]$$
$$a_{ij} = \exp \left(-\frac{d_{ij}^2}{\sigma^2} \right), \text{ when } i \neq j, \exp \left(-\frac{d_{ij}^2}{\sigma^2} \right) \geq \epsilon$$

where ϕ^i, λ^i is the latitude, longitude of sensor i , with σ^2, ϵ controlling the sparsity of A , and R represents Earth's radius.

E-STGCN: Spatial Module

- Encode the structural and feature-based aspects of the graph using GCNs (K layers) through neural message passing mechanism for node i as follows:

$$h_t^{i,(0)} = X_t^i$$
$$h_t^{i,(k)} = f_t^{(k)} \left(W_t^{(k)} \frac{\sum_{j \in \mathcal{N}(i)} h_t^{j,(k-1)}}{|\mathcal{N}(i)|} + B_t^{(k)} h_t^{i,(k-1)} \right) ; k = 1, 2, \dots, K$$

- Generate the spatiotemporal embedding based on the GCNs representations using an MLP as:

$$Z_t^i = \text{Dense} \left(h_t^{i,(K)} \right)$$

E-STGCN: Temporal Module

- Provide the spatiotemporal embedding $\mathbf{Z}_t = \{Z_t^1, \dots, Z_t^N\}$ to the [Temporal Block: LSTM layers](#) and a [Dense layer](#) to generate the desired forecasts.
- The mathematical formulation of the chain-like architecture in LSTM with input gate ($\mathbf{I}_t$), forget gate ($\mathbf{F}_t$), and output gate ($\mathbf{O}_t$) can be expressed as:

$$\begin{aligned}\mathbf{F}_t &= \text{ReLU}(U^F \mathbf{Y}_t + B^F); \mathbf{I}_t = \text{ReLU}(U^I \mathbf{Y}_t + B^I); \\ \mathbf{M}_t &= \tanh(U^M \mathbf{Y}_t + B^M); \mathbf{C}_t = (\mathbf{F}_t \odot \mathbf{C}_{t-1} \oplus \mathbf{I}_t \odot \mathbf{M}_t); \\ \mathbf{O}_t &= \text{ReLU}(U^O \mathbf{Y}_t + B^O); \mathbf{H}_t = (\mathbf{O}_t \odot \tanh(\mathbf{C}_t))\end{aligned}$$

where $\mathbf{Y}_t$ is the concatenation matrix with p lagged values of $\mathbf{Z}_t$ and previous hidden state $\mathbf{H}_{t-1}$, the new memory $\mathbf{M}_t$ at time t is used to update the cell state $\mathbf{C}_t$ (long term memory) and hidden state $\mathbf{H}_t$ (short term memory).

- Based on the computed values of $\mathbf{C}_t$ and $\mathbf{H}_t$ the q steps ahead forecast of $\mathbf{Z}_t$ is computed as:

$$\{\hat{\mathbf{Z}}_{t+1}, \hat{\mathbf{Z}}_{t+2}, \dots, \hat{\mathbf{Z}}_{t+q}\} = \text{ReLU}(U^H \mathbf{H}_t + B^H)$$

with U and B represents the recurrent weights matrix and the bias vector to be optimized based on the [proposed loss function](#).

E-STGCN: EVT Module

- EVT module performs PoT approach on the pollutant data.
- Hence, the conditional GPD for the i^{th} monitoring station at time t is:

$$P[X_t^i - \tau \leq x_t^i \mid X_t^i \geq \tau] = \begin{cases} 1 - \left(1 + \frac{\xi^i x_t^i}{\alpha^i}\right)^{-1/\xi_t^i}, & \xi^i \neq 0 \\ 1 - \exp\left(-\frac{x_t^i}{\alpha^i}\right)^2, & \xi^i = 0 \end{cases}$$

where $\xi^i (\in \mathbb{R})$ is the shape parameter and $\alpha^i (> 0)$ is the scale parameter.

- The POTL loss function is the deviance or **negative log-likelihood** loss from GPD

$$PoTL(z_t^i) = \text{NLL}(\text{GPD}) = \log g(\hat{\alpha}^i) + \left(1 + \frac{1}{\hat{\xi}^i}\right) \log g\left(1 + \frac{\hat{\xi}^i z_t^i}{\hat{\alpha}^i}\right)$$

for location i for one data point. **Lower NLL = Better Model Fit**

E-STGCN: Optimization

- Optimize the network using the following loss function (based on the **data loss** and the **PoT-based loss**):

$$\text{Loss}(Z_{t+q}^i, \hat{Z}_{t+q}^i) = \begin{cases} \text{MSE}(Z_{t+q}^i, \hat{Z}_{t+q}^i), & \hat{Z}_{t+q}^i \leq \tau, \\ \beta_1 \text{MSE}(Z_{t+q}^i, \hat{Z}_{t+q}^i) + \beta_2 \text{POTL}(\hat{Z}_{t+q}^i), & \hat{Z}_{t+q}^i > \tau, \end{cases}$$

Hyperparameters
(cross validation)

- Following PINN, the final loss function of the LSTM network is formulated as a combination of the data loss and the **POT-based loss function whenever the predicted values exceed the NAAQS threshold τ** .
- Since the loss function is differentiable almost everywhere, we employ the backpropagation approach to train the corresponding weights.
- The modified loss function employed for handling the temporal dependency in the E-STGCN framework integrates prior distributional knowledge to the predictions in case of threshold exceedances.

Statistical Significance

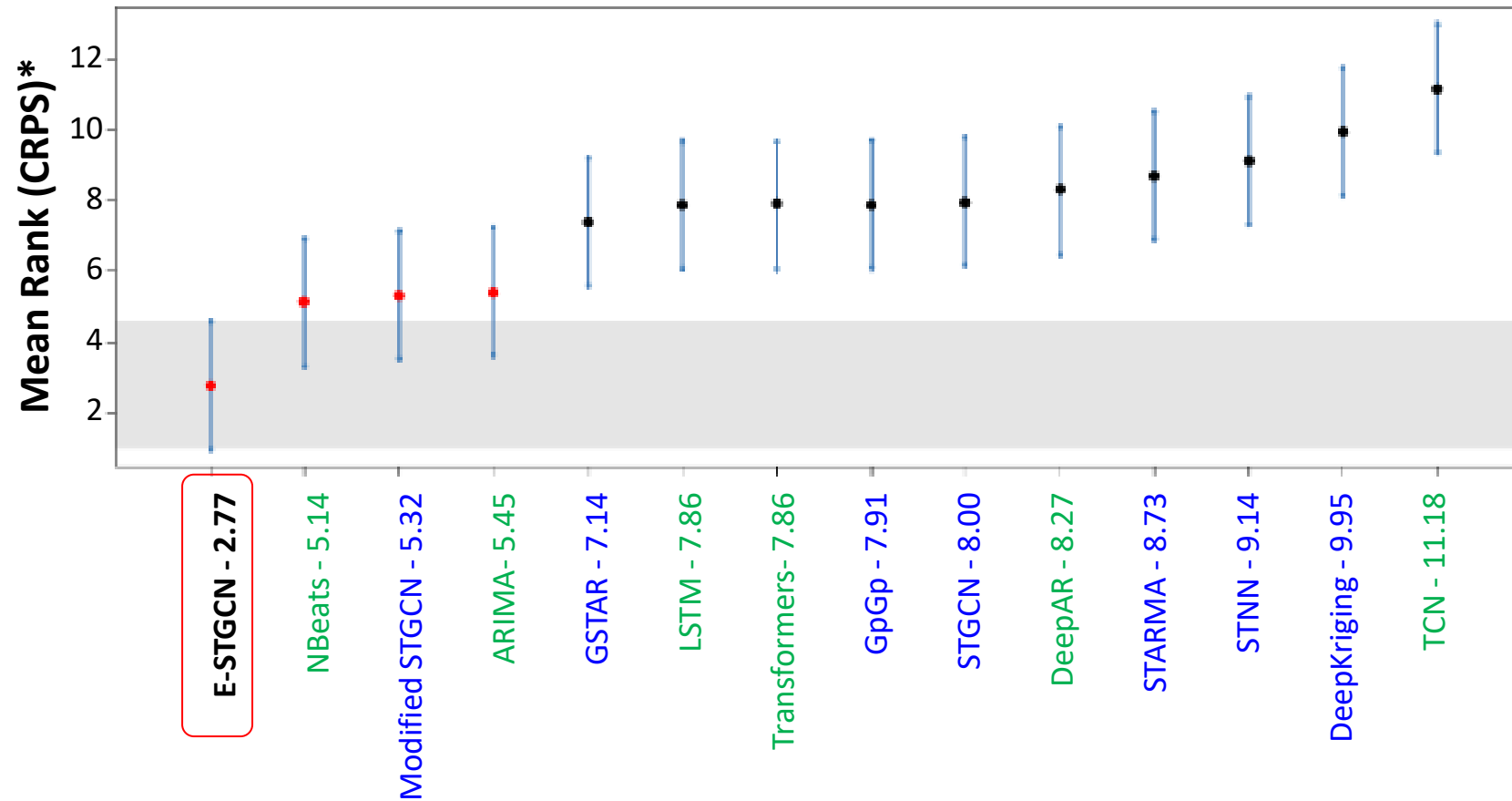
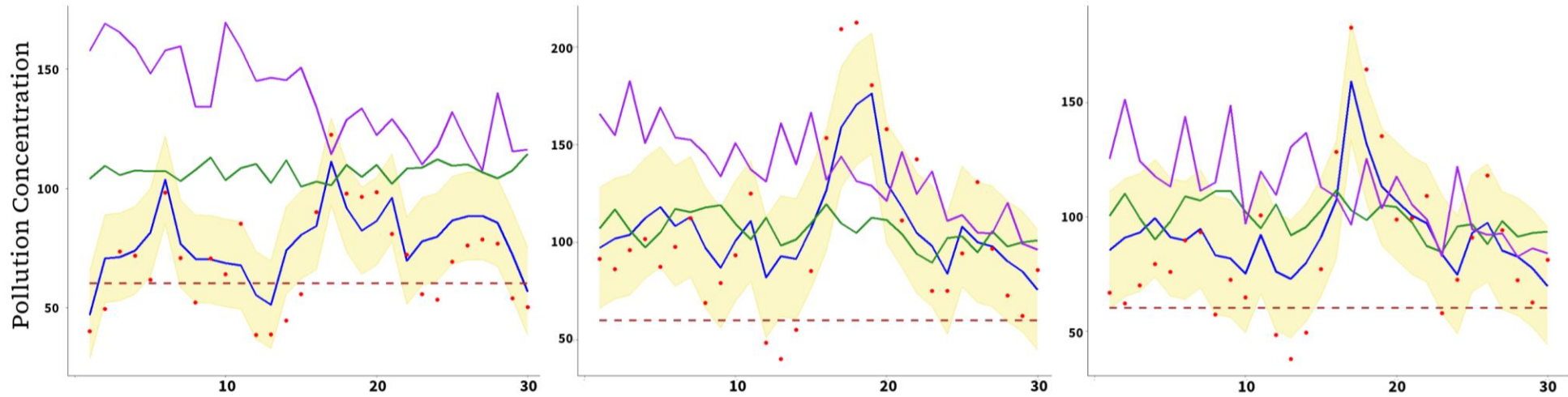


Fig: MCB test comparing the performance of E-STGCN with **spatiotemporal** and **temporal** models for PM_{2.5} forecasting based on CRPS metric

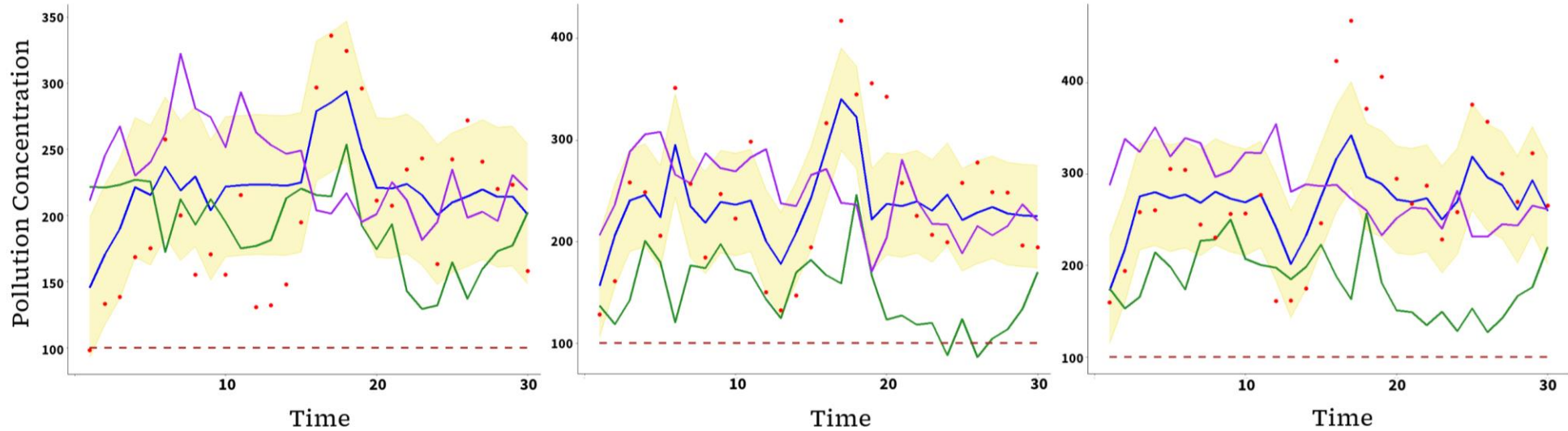
*Continuous Ranked Probability Score (CRPS): Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *JASA*.

E-STGCN Prediction (Feb 2023)

$\text{PM}_{2.5}$



PM_{10}

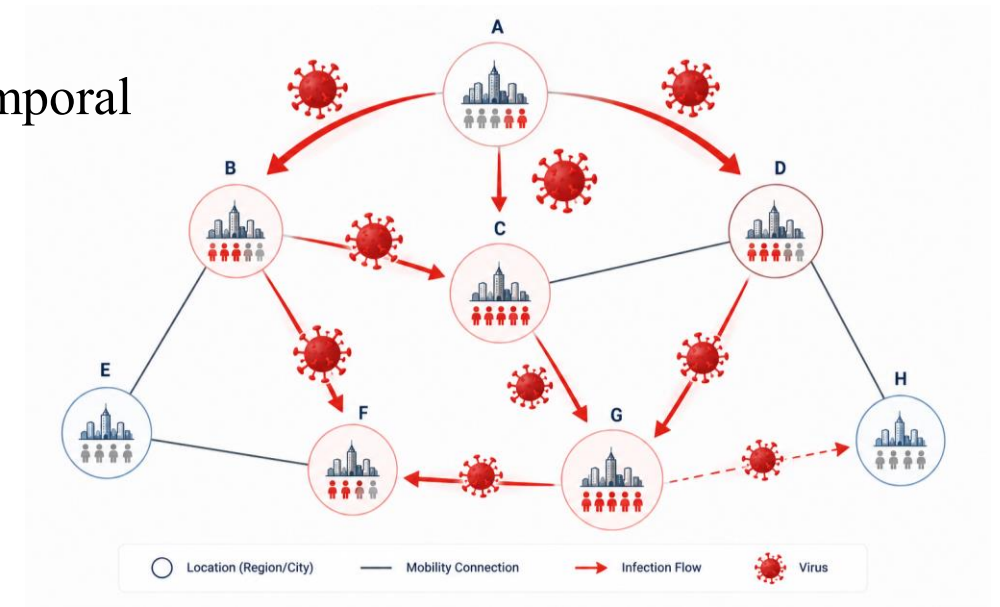


Ground Truth E-STGCN Modified STGCN NBeats Prediction Interval NAAQS Threshold

*Xu, Chen, and Yao Xie. "Conformal prediction interval for dynamic time-series." *ICML* 2021.

Remark

- E-STGCN combined geometric deep-learning architecture and extreme value theory
- Data-centric model that doesn't include physical knowledge
- Outperforms for air quality forecasting, where dense, high-frequency sensor observations are available
- Epidemic datasets are sparse, limited in size, and are spatiotemporal
- Probabilistic forecasts quantify uncertainty, which is critical for risk assessment and managing outbreak events
- Conformal prediction is a model-agnostic/post-hoc rather than model-intrinsic. Its main limitation is that it provides calibrated prediction sets but does not learn the underlying conditional distribution and GP has computational complexity.



Probabilistic Spatiotemporal Forecasting

Published in Transactions on Machine Learning Research (06/2026)



Deep Generative Spatiotemporal Engression for Probabilistic Forecasting of Epidemics

Rajdeep Pathak

Tanujit Chakraborty

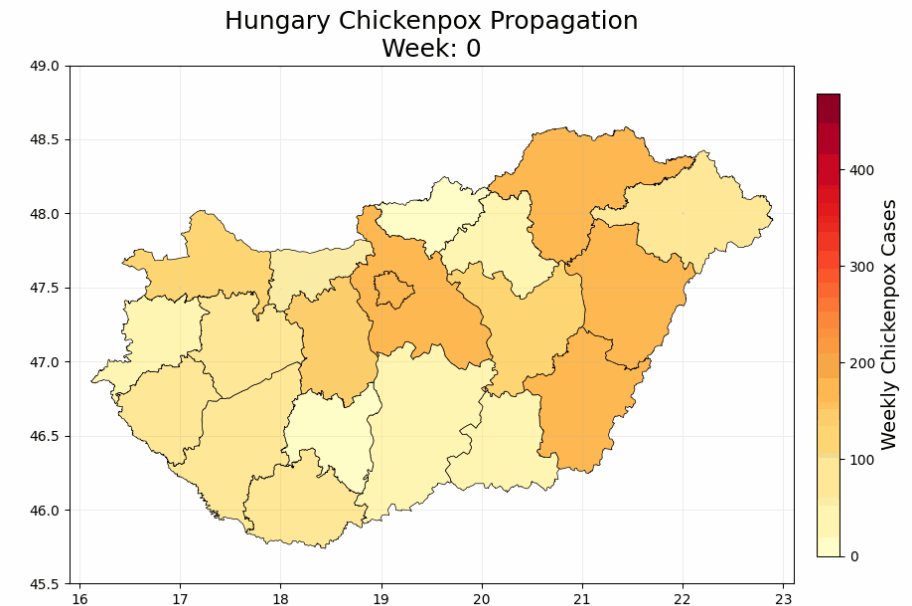
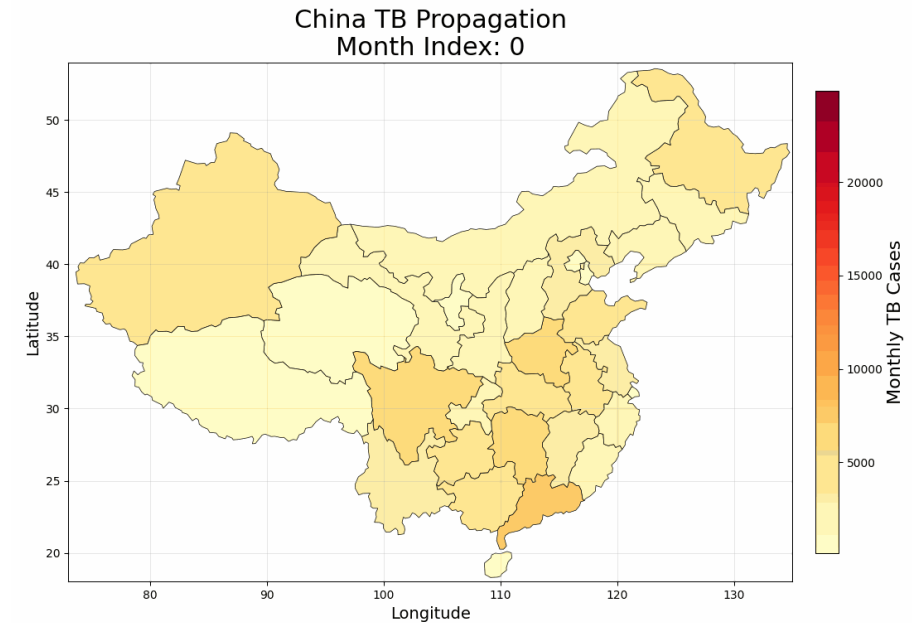
SAFIR, Sorbonne University Abu Dhabi, United Arab Emirates

SCAI, Sorbonne Université, Paris, France



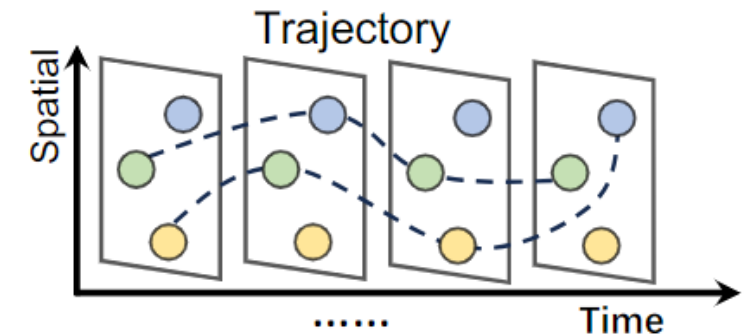
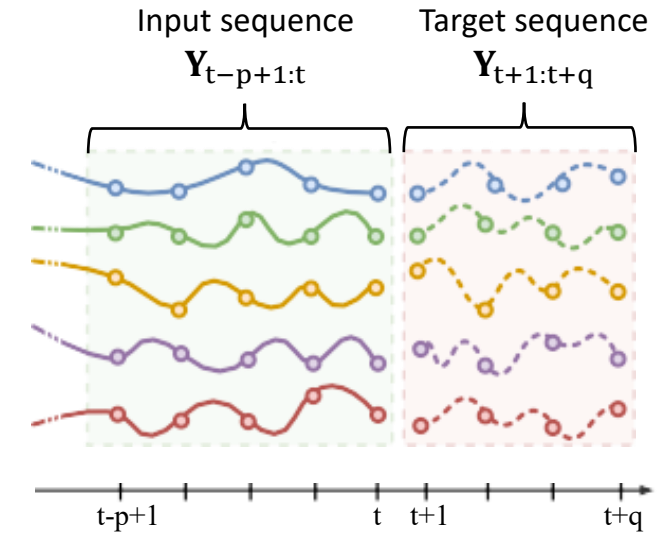
Epidemic Datasets

- Accurate forecasting of epidemics is necessary for:
 - Timely deployment of control measures
 - Allocation of medical resources
 - Formulation of intervention strategies
- Point forecasts have certain limitations
- **Uncertainty quantification:** Probabilistic forecasts quantify uncertainty, which is critical for risk assessment and managing extreme outbreak events



Engression: The Concept

- Engression is a generative deep distributional regression framework introduced by [Shen and Meinshausen \(2025\)](#).
- **Goal:** Estimate the full conditional distribution $P(Y|X = x)$ rather than just the conditional mean.
- **Pre-additive Noise Model (pre-ANM):** Unlike standard regression $Y = f(X) + \eta$, engression adds noise *before* the non-linear transformation:
$$Y = g(X + \eta)$$
- Engression model is extrapolative.



Graph Convolutional Engression Network

- GCEN combines Graph Convolutional Networks (GCNs) and Long Short-Term Memory (LSTM) with Engression.
- **Spatial Feature Extraction (GCN):** Extracts spatially-aware embeddings Z_τ using a weighted adjacency matrix A based on Haversine distance.

$$Z_\tau = f_{GCN}(Y_\tau, A; \Theta_{GCN})$$

- **Latent Noise Injection:** Random noise η_τ is added to the spatial embeddings:

$$Z'_\tau = Z_\tau + \eta_\tau$$

- **Temporal Evolution (LSTM):** A LSTM module followed by a fully connected layer predicts future states based on perturbed embeddings:

$$\mathcal{H}_\tau = LSTM(Z'_\tau, \mathcal{H}_{\tau-1})$$

GCEN: Noise Injection and Temporal Module

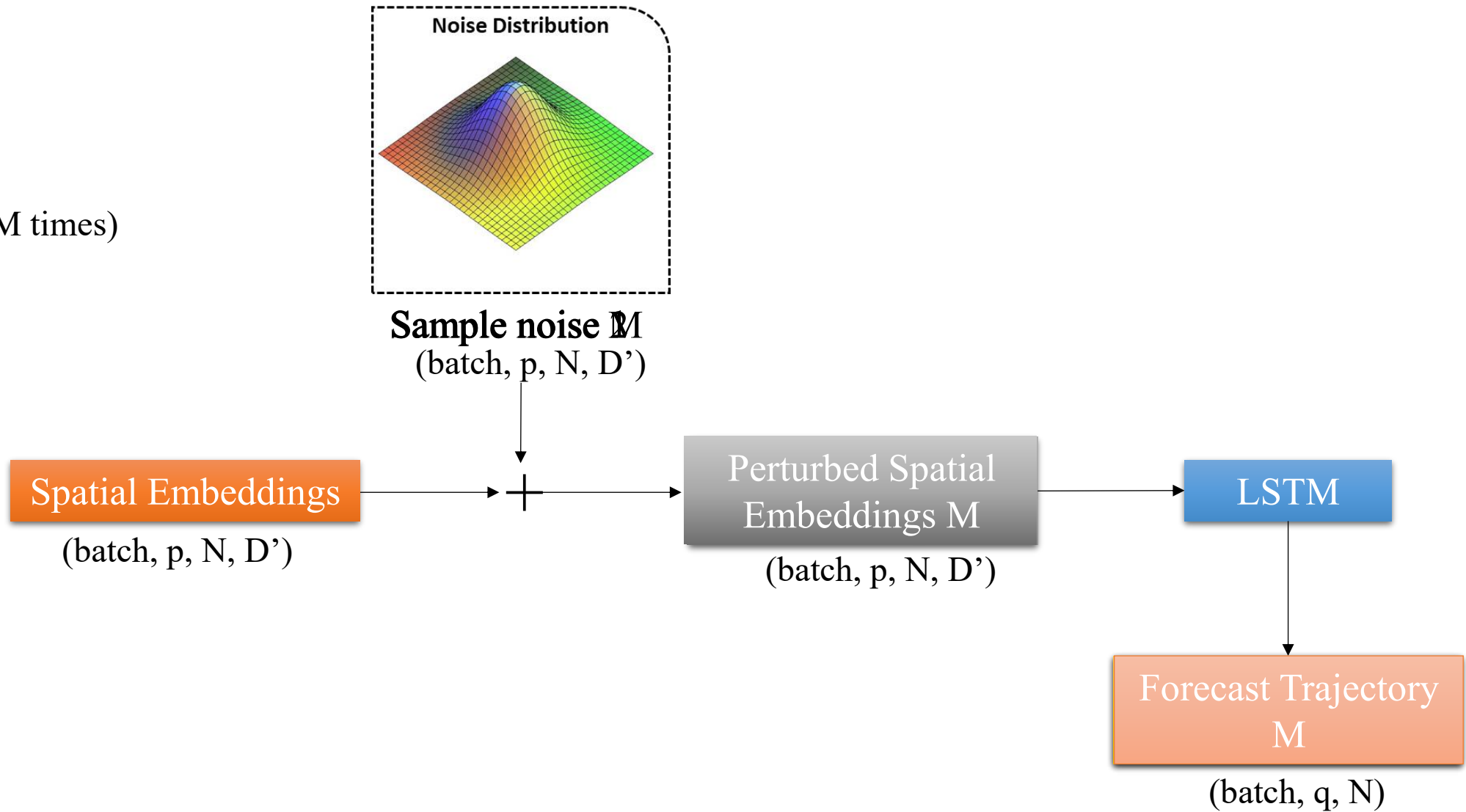
- Generate a random noise from a pre-determined distribution (e.g., standard Gaussian or Uniform) and add it to the spatiotemporal embeddings:

$$Z_t^i = Z_t^i + \eta_t^i$$

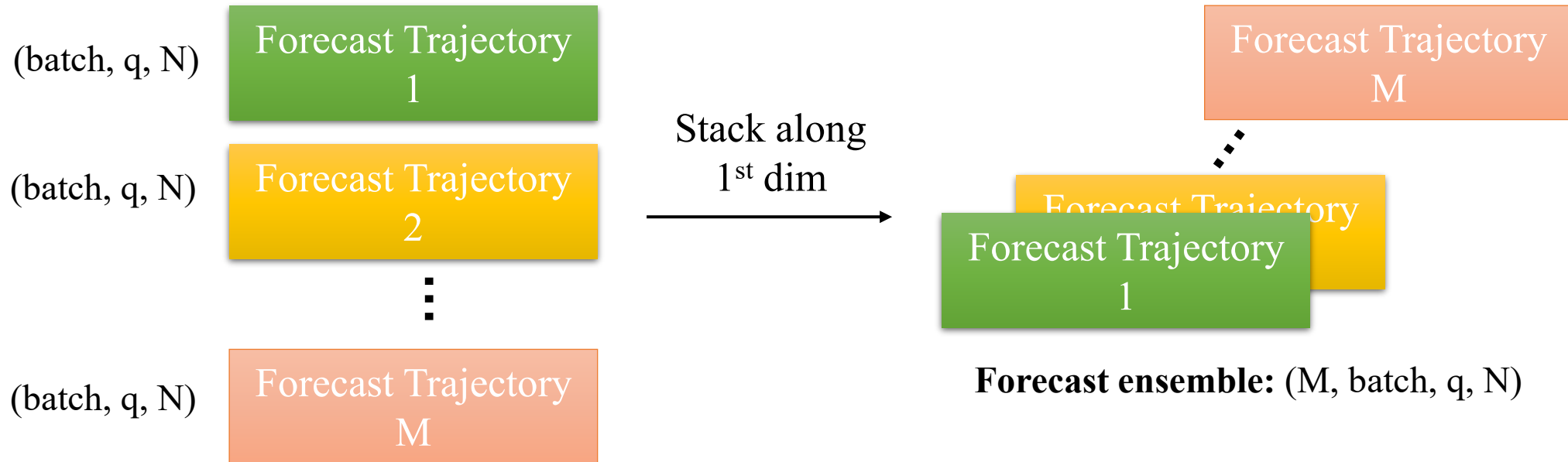
- For each time-step, sample M independent noise values.
- Provide the M different perturbed spatiotemporal embeddings to the temporal block: LSTM layers followed by a fully connected layer.
- The result is a forecast ensemble: M different plausible forecasts for the desired forecast horizon.

GCEN: An Illustration

(repeat for M times)



GCEN: Forecast Ensemble Formation



Optimization: Energy Score Loss

- Training dataset: $\mathcal{D} = \{(\mathbf{Y}_{t-p+1:t}, \mathbf{Y}_{t+1:t+q})\}, p \leq t \leq T - q$
- The models are trained using the energy score loss:

$$\mathcal{L}_{ES} = \frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \left[\underbrace{\frac{1}{M} \sum_{j=1}^M \left\| \mathbf{Y}_i - \hat{\mathbf{Y}}_i^{(j)} \right\|_2^\beta}_{\text{Accuracy term (pull samples towards ground truth)}} - \underbrace{\frac{1}{2M(M-1)} \sum_{j=1}^M \sum_{k=1}^M \left\| \hat{\mathbf{Y}}_i^{(j)} - \hat{\mathbf{Y}}_i^{(k)} \right\|_2^\beta}_{\text{Sharpness/dispersion term (encourage variance among samples)}} \right]$$

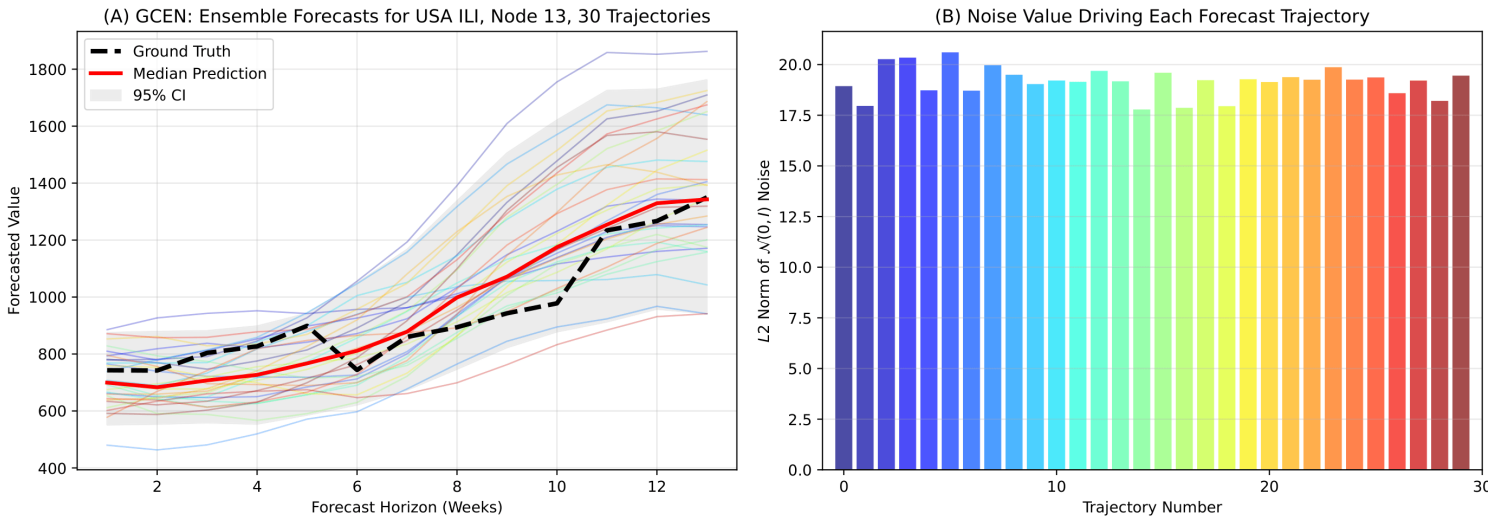
where, $\mathbf{Y}_i$ is the ground truth sequence of length q , and $\hat{\mathbf{Y}}_i^{(j)}$ is its predicted value corresponding to the noise $\eta_i^{(j)}$

- M is the number of predictive samples generated for each training example and β controls how strongly larger forecast errors or sample separations are penalized (0, 2).

Generative Forecasting Process

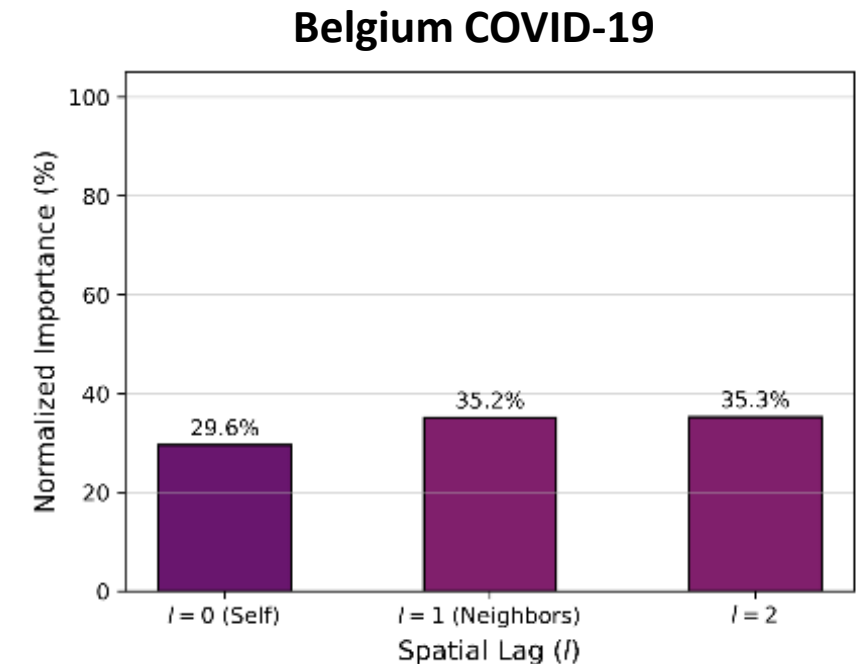
- GCEN is a generative sequence-to-sequence model.
- To produce a probabilistic forecast:
 - ✓ Input historical sequence $Y_{t-p+1:t}$
 - ✓ Generate spatial embeddings $Z_{t-p+1:t}$
 - ✓ Sample M independent noise vectors $\{\eta^{(1)}, \dots, \eta^{(M)}\}$
 - ✓ Perturb embeddings and pass through the LSTM temporal module M times
 - ✓ **Output:** An ensemble of M distinct future trajectories
- This ensemble allows for the construction of model-intrinsic prediction intervals and risk assessment.
- Theoretically, when formulated as an autoregressive 1-step-recurrent Markov chain, the chains representing the GCEN process IS **geometrically ergodic and asymptotically stationary** under some mild assumptions.

Model Explainability

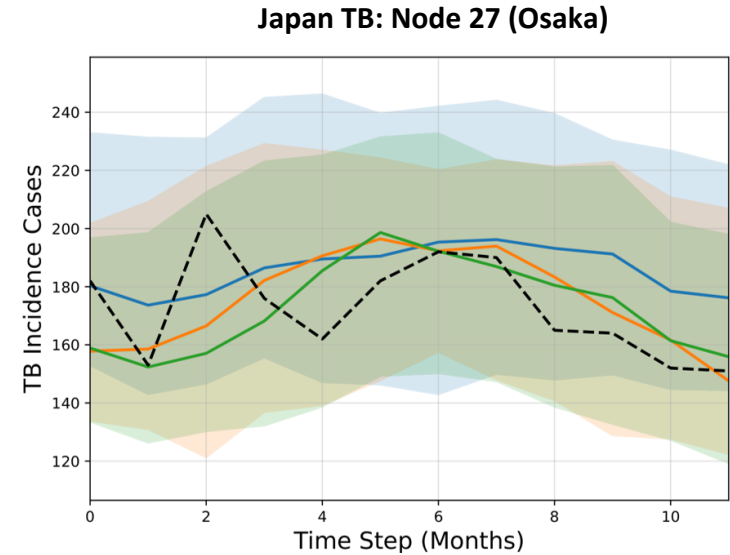
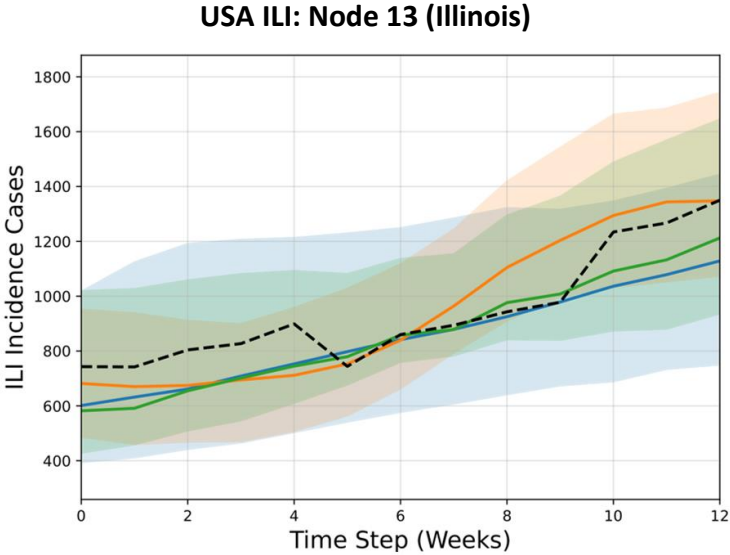
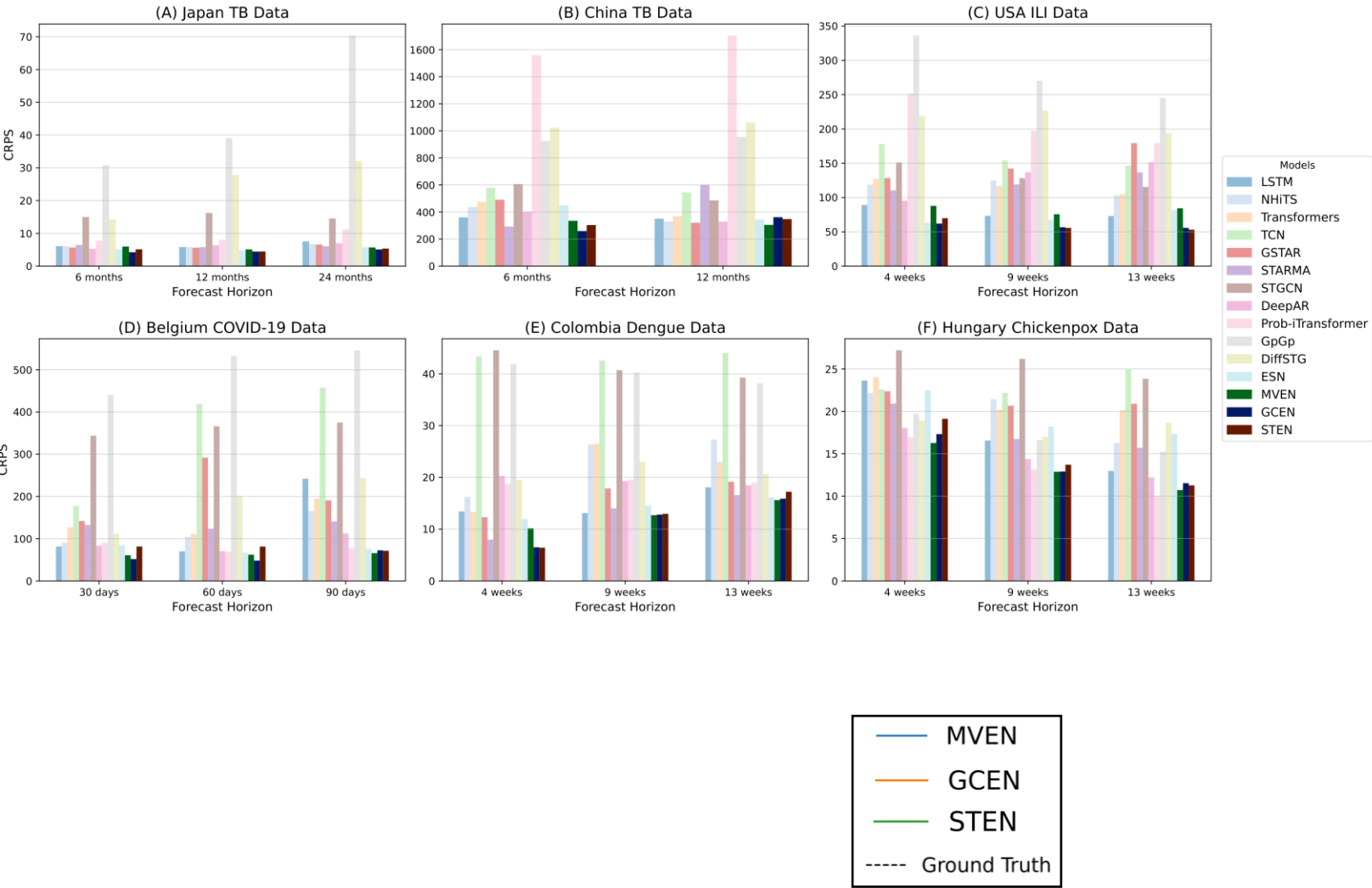


For node 13 of USA ILI data, this visualizes the norm of the noise values that drives the 50 forecast trajectories.

- Learned importance of spatial lags by STEN.
- E.g., STEN learned that Belgium COVID-19 exhibits a clear spatial diffusion profile, where the influence of first-order ($l = 1$, 35.2%) and second-order neighbors ($l = 2$, 35.3%) outweighs local history ($l = 0$, 29.6%).



Results: CRPS and Example Forecasts

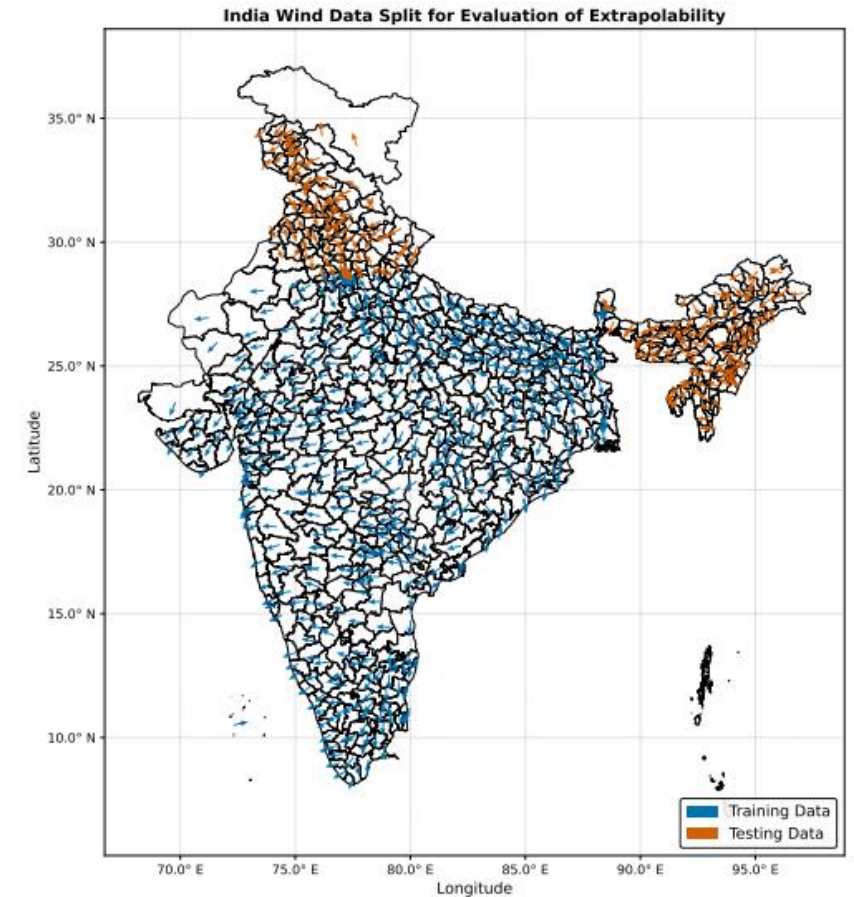


Future Direction

Future Work

Probabilistic Learning for Circular + Spatiotemporal Data

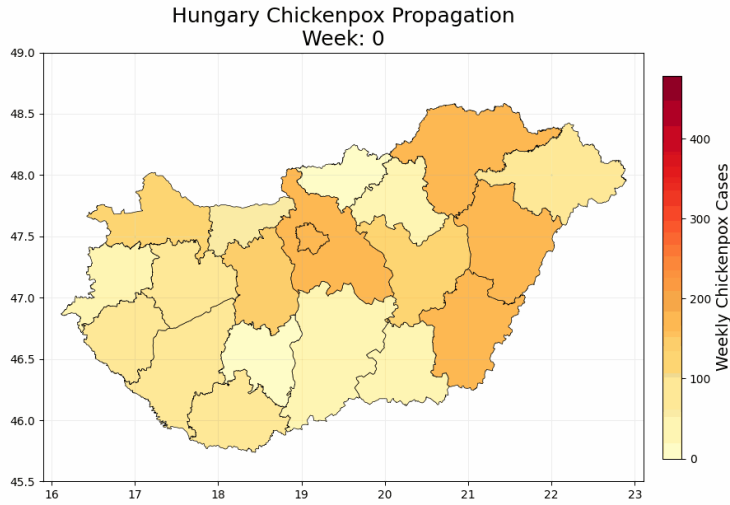
- Circular data refers to observations that take the form of angles or directions on S^1
- Interesting applications: Object pose estimation, gaze direction prediction, wind direction, etc.
- Alzheimer's patients involuntary movement prediction.



Research Areas and current PhD students at SU



Rajdeep Pathak
(Nov 2025 - Ongoing)
ED-386, SU Paris
Funded by SAFIR, SUAD



Spatiotemporal AI



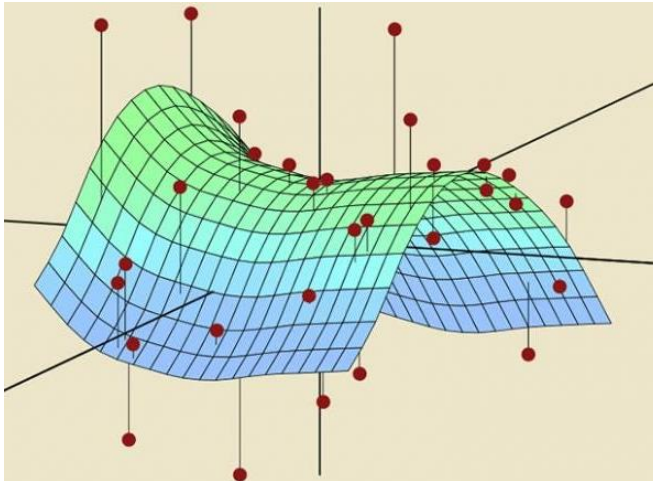
Ali Rammal
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AI in Health



Yueyan Zhang
(Just Joined)
ED-386, SU Paris
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Statistical Learning Theory



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Funded by SOUND.AI



Generative AI

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essentially,
all models are wrong,
but some are useful

George E. P. Box

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